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import React, { useState, useEffect, useRef } from 'react';
import { Volume2, Mic, Send, Settings, BookOpen, MessageCircle, RotateCw, AlertCi

const EnglishLearningSaaS = () => {
  const [currentStructureIdx, setCurrentStructureIdx] = useState(0);
  const [selectedPhrase, setSelectedPhrase] = useState(null);
  const [conversationMode, setConversationMode] = useState('structure'); // 'stru
  const [messages, setMessages] = useState([]);
  const [userInput, setUserInput] = useState('');
  const [isListening, setIsListening] = useState(false);
  const [correction, setCorrection] = useState(null);
  const [pronunciationLang, setPronunciationLang] = useState('en-US');
  const [showExplanation, setShowExplanation] = useState(false);
  const recognitionRef = useRef(null);

  const STRUCTURES = [
    {
      structure: "I need",
      explanation: "Exprimer un besoin ou une demande polite",
      examples: [
        "I need help with this problem",
        "I need more time to finish",
        "I need a break right now",
        "I need your opinion on this",
        "I need to speak with you",
        "I need to reschedule the meeting",
        "I need better internet connection",
        "I need information about the project"
      ]
    },
    {
      structure: "Would you like",
      explanation: "Proposer quelque chose poliment ou demander une préférence",
      examples: [
        "Would you like to join our team meeting?",
        "Would you like a coffee or tea?",
        "Would you like to see the presentation?",
        "Would you like to work remotely today?",
        "Would you like me to send the files?",
        "Would you like to discuss this later?",
        "Would you like a detailed explanation?",
        "Would you like to schedule a call?"
      ]
    }
  ],

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{
  structure: "Can you",
  explanation: "Demander si quelqu'un peut faire quelque chose (compétence ou
examples: [
  "Can you send me the report?",
  "Can you help me with this task?",
  "Can you explain this concept?",
  "Can you attend the meeting tomorrow?",
  "Can you review this document?",
  "Can you provide more details?",
  "Can you fix this issue?",
  "Can you speak English fluently?"
]
},
{
  structure: "I should",
  explanation: "Exprimer un conseil, une obligation ou une recommandation",
  examples: [
    "I should finish this before Friday",
    "I should improve my English skills",
    "I should take a day off",
    "I should read more technical documents",
    "I should practice speaking daily",
    "I should update the database",
    "I should contact the client soon",
    "I should learn new programming languages"
  ]
},
{
  structure: "Could you please",
  explanation: "Faire une demande très polite et formelle",
  examples: [
    "Could you please review this proposal?",
    "Could you please send the invoice?",
    "Could you please clarify this point?",
    "Could you please share your thoughts?",
    "Could you please wait a moment?",
    "Could you please repeat that?",
    "Could you please provide evidence?",
    "Could you please sign this document?"
  ]
},
{
  structure: "I think that",
  explanation: "Donner son opinion ou exprimer une pensée",
  examples: [
    "I think that this approach is better",

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    "I think that we should try again",
    "I think that the project is progressing well",
    "I think that collaboration is important",
    "I think that this deadline is unrealistic",
    "I think that we need more resources",
    "I think that the quality is excellent",
    "I think that this solution won't work"
  ]
},
{
  structure: "It seems to me",
  explanation: "Donner une opinion de manière moins directe et plus modeste",
  examples: [
    "It seems to me that the plan is solid",
    "It seems to me we're running out of time",
    "It seems to me the client is satisfied",
    "It seems to me this is a priority",
    "It seems to me everyone agrees",
    "It seems to me the budget is tight",
    "It seems to me the design is modern",
    "It seems to me the team is motivated"
  ]
},
{
  structure: "As far as I know",
  explanation: "Partager une information avec certitude limitée",
  examples: [
    "As far as I know, the meeting is next week",
    "As far as I know, she's still working here",
    "As far as I know, the project started yesterday",
    "As far as I know, that's the correct procedure",
    "As far as I know, no one has objected",
    "As far as I know, the system is working",
    "As far as I know, he has experience",
    "As far as I know, it's already approved"
  ]
},
{
  structure: "According to",
  explanation: "Citer une source ou référencer une information",
  examples: [
    "According to the report, sales increased",
    "According to the manager, the deadline is flexible",
    "According to statistics, productivity improved",
    "According to the documentation, the process is clear",
    "According to my research, this is accurate",
    "According to the client, quality is essential",
  ]
}

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        "According to the data, we're on track",
        "According to the guidelines, this is correct"
    ]
},
{
    structure: "In my opinion",
    explanation: "Exprimer clairement son avis ou sa perspective personnelle",
    examples: [
        "In my opinion, this is the best solution",
        "In my opinion, we should invest more",
        "In my opinion, the timing is perfect",
        "In my opinion, communication is key",
        "In my opinion, the price is reasonable",
        "In my opinion, innovation matters",
        "In my opinion, the team deserves recognition",
        "In my opinion, this strategy is effective"
    ]
},
{
    structure: "I must say",
    explanation: "Introduire une observation ou un commentaire important",
    examples: [
        "I must say, the results are impressive",
        "I must say, this is challenging",
        "I must say, the team performed well",
        "I must say, I'm impressed by your work",
        "I must say, this is unexpected",
        "I must say, the presentation was excellent",
        "I must say, the progress is remarkable",
        "I must say, I have concerns about this"
    ]
},
{
    structure: "Let me",
    explanation: "Proposer ou offrir de faire quelque chose",
    examples: [
        "Let me help you with this task",
        "Let me explain the situation",
        "Let me show you the results",
        "Let me know if you need anything",
        "Let me call you back later",
        "Let me think about this",
        "Let me send you the files",
        "Let me check the schedule"
    ]
},
{

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    structure: "I'm not sure if",
    explanation: "Exprimer de l'incertitude ou du doute",
    examples: [
        "I'm not sure if this will work",
        "I'm not sure if he'll attend",
        "I'm not sure if the deadline is realistic",
        "I'm not sure if we have the budget",
        "I'm not sure if that's the right approach",
        "I'm not sure if she has the experience",
        "I'm not sure if it's available now",
        "I'm not sure if the data is accurate"
    ]
},
{
    structure: "It's important to",
    explanation: "Souligner l'importance d'une action ou d'une condition",
    examples: [
        "It's important to meet deadlines",
        "It's important to communicate clearly",
        "It's important to follow the guidelines",
        "It's important to maintain quality",
        "It's important to document everything",
        "It's important to test thoroughly",
        "It's important to listen to feedback",
        "It's important to prioritize tasks"
    ]
},
{
    structure: "I suggest",
    explanation: "Proposer une idée ou une action à quelqu'un",
    examples: [
        "I suggest we schedule a meeting",
        "I suggest a different approach",
        "I suggest starting from scratch",
        "I suggest reviewing the documentation",
        "I suggest waiting for more data",
        "I suggest postponing the decision",
        "I suggest hiring an expert",
        "I suggest investing in training"
    ]
},
{
    structure: "I'm confident that",
    explanation: "Exprimer de la confiance ou de la certitude sur quelque chose",
    examples: [
        "I'm confident that this will succeed",
        "I'm confident that the team is capable",

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        "I'm confident that we'll meet the target",
        "I'm confident that the plan is solid",
        "I'm confident that quality will be excellent",
        "I'm confident that we made the right choice",
        "I'm confident that the solution works",
        "I'm confident that the data is reliable"
    ]
},
{
    structure: "I'll definitely",
    explanation: "Promesse ferme ou confirmation d'une action future",
    examples: [
        "I'll definitely attend the meeting",
        "I'll definitely send the report",
        "I'll definitely follow up on this",
        "I'll definitely improve my skills",
        "I'll definitely keep you updated",
        "I'll definitely deliver on time",
        "I'll definitely consider your feedback",
        "I'll definitely check the details"
    ]
},
{
    structure: "It would be better if",
    explanation: "Suggérer une amélioration de manière modérée",
    examples: [
        "It would be better if we started earlier",
        "It would be better if we had more resources",
        "It would be better if the deadline was flexible",
        "It would be better if communication was clearer",
        "It would be better if we tested more",
        "It would be better if we had feedback",
        "It would be better if the team was larger",
        "It would be better if we planned better"
    ]
},
{
    structure: "On the other hand",
    explanation: "Présenter un argument contraire ou un point de vue différent"
    examples: [
        "On the other hand, cost is a factor",
        "On the other hand, we lack experience",
        "On the other hand, time is limited",
        "On the other hand, the market is unstable",
        "On the other hand, competition is fierce",
        "On the other hand, quality might suffer",
        "On the other hand, it could be risky",
    ]
}

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    "On the other hand, the risk is high"
  ]
},
{
  structure: "I realize that",
  explanation: "Reconnaître ou prendre conscience de quelque chose",
  examples: [
    "I realize that I made a mistake",
    "I realize that this is complicated",
    "I realize that we need more planning",
    "I realize that communication was poor",
    "I realize that the deadline is tomorrow",
    "I realize that I misunderstood",
    "I realize that this requires expertise",
    "I realize that the problem is bigger"
  ]
},
{
  structure: "It turns out that",
  explanation: "Révéler un résultat inattendu ou une découverte",
  examples: [
    "It turns out that the system has a bug",
    "It turns out that she has more experience",
    "It turns out that the cost is higher",
    "It turns out that everyone agrees",
    "It turns out that the data was wrong",
    "It turns out that it's more difficult",
    "It turns out that he can't attend",
    "It turns out that we need new equipment"
  ]
},
{
  structure: "I'd like to point out",
  explanation: "Attirer l'attention sur quelque chose d'important",
  examples: [
    "I'd like to point out that this is urgent",
    "I'd like to point out that we're ahead",
    "I'd like to point out that this was overlooked",
    "I'd like to point out that the quality improved",
    "I'd like to point out that costs increased",
    "I'd like to point out that he's qualified",
    "I'd like to point out that we agreed",
    "I'd like to point out that the risk exists"
  ]
},
{
  structure: "To put it simply",

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    explanation: "Simplifier une explication complexe",
    examples: [
      "To put it simply, we need more money",
      "To put it simply, the plan won't work",
      "To put it simply, we're running late",
      "To put it simply, it's too expensive",
      "To put it simply, we're understaffed",
      "To put it simply, it's a bad idea",
      "To put it simply, we succeeded",
      "To put it simply, quality matters most"
    ]
  },
  {
    structure: "What I mean is",
    explanation: "Clarifier ou expliquer ce qu'on vient de dire",
    examples: [
      "What I mean is that timing is critical",
      "What I mean is we need better planning",
      "What I mean is communication is essential",
      "What I mean is the budget is tight",
      "What I mean is the deadline is fixed",
      "What I mean is collaboration works best",
      "What I mean is quality is non-negotiable",
      "What I mean is we need more training"
    ]
  },
  {
    structure: "To be honest",
    explanation: "Introduire une opinion franche et sincère",
    examples: [
      "To be honest, I don't think this works",
      "To be honest, I'm concerned about this",
      "To be honest, the results are disappointing",
      "To be honest, I prefer the other option",
      "To be honest, I have doubts",
      "To be honest, the quality needs improvement",
      "To be honest, I'm not confident",
      "To be honest, we're not ready yet"
    ]
  }
];

// Initialize speech recognition
useEffect(() => {
  const SpeechRecognition = window.SpeechRecognition || window.webkitSpeechRecognition
  if (SpeechRecognition) {
    recognitionRef.current = new SpeechRecognition();
  }
});

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recognitionRef.current.continuous = false;
recognitionRef.current.interimResults = false;

recognitionRef.current.onresult = (event) => {
  const transcript = Array.from(event.results)
    .map(result => result[0].transcript)
    .join('');

  handleVoiceInput(transcript);
  setIsListening(false);
};

recognitionRef.current.onerror = (event) => {
  console.error('Speech recognition error:', event.error);
  setIsListening(false);
};

recognitionRef.current.onend = () => {
  setIsListening(false);
};
}
}, []));

const handleVoiceInput = (transcript) => {
  if (conversationMode === 'structure' && selectedPhrase) {
    const originalPhrase = STRUCTURES[currentStructureIdx].examples[selectedPhr
    const similarity = calculateSimilarity(transcript.toLowerCase(), originalPh

    if (similarity > 0.7) {
      setCorrection({
        type: 'success',
        message: `✓ Excellent! You said: "${transcript}"`,
        percentage: Math.round(similarity * 100)
      });
    } else {
      setCorrection({
        type: 'error',
        message: `You said: "${transcript}"\nCorrect: "${originalPhrase}"`,
        percentage: Math.round(similarity * 100)
      });
    }
  } else {
    setUserInput(transcript);
  }
};

const calculateSimilarity = (str1, str2) => {

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const longer = str1.length > str2.length ? str1 : str2;
const shorter = str1.length > str2.length ? str2 : str1;
if (longer.length === 0) return 1.0;

const editDistance = getEditDistance(longer, shorter);
return (longer.length - editDistance) / longer.length;
};

const getEditDistance = (s1, s2) => {
  const costs = [];
  for (let i = 0; i <= s1.length; i++) {
    let lastValue = i;
    for (let j = 0; j <= s2.length; j++) {
      if (i === 0) {
        costs[j] = j;
      } else if (j > 0) {
        let newValue = costs[j - 1];
        if (s1.charAt(i - 1) !== s2.charAt(j - 1)) {
          newValue = Math.min(Math.min(newValue, lastValue), costs[j]) + 1;
        }
        costs[j - 1] = lastValue;
        lastValue = newValue;
      }
    }
    if (i > 0) costs[s2.length] = lastValue;
  }
  return costs[s2.length];
};

const speak = (text, lang = pronunciationLang) => {
  const utterance = new SpeechSynthesisUtterance(text);
  utterance.lang = lang;
  utterance.rate = 0.9;
  window.speechSynthesis.speak(utterance);
};

const startListening = () => {
  if (recognitionRef.current) {
    recognitionRef.current.lang = pronunciationLang;
    recognitionRef.current.start();
    setIsListening(true);
    setCorrection(null);
  }
};

const sendChatMessage = async () => {
  if (!userInput.trim()) return;

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const newMessages = [...messages, { role: 'user', content: userInput }];
setMessages(newMessages);
setUserInput('');

// Simulate AI response
const responses = [
  "That's a great point! Can you explain more?",
  "I understand. What would you like to discuss next?",
  "Interesting perspective. Tell me more about this.",
  "Good question! Let me help you with that.",
  "I appreciate your input. How can I assist further?"
];

setTimeout(() => {
  const response = responses[Math.floor(Math.random() * responses.length)];
  setMessages([...newMessages, { role: 'assistant', content: response }]);
}, 500);
};

const handlePhraseSelection = (idx) => {
  setSelectedPhrase(selectedPhrase === idx ? null : idx);
  setCorrection(null);
};

const currentStructure = STRUCTURES[currentStructureIdx];

return (
  <div className="min-h-screen bg-gradient-to-br from-blue-50 via-white to-indigo-50" /* Header */>
    <div className="sticky top-0 z-40 bg-white/95 backdrop-blur-sm border-b border-gray-200">
      <div className="max-w-6xl mx-auto px-4 py-4 flex justify-between items-center">
        <div className="flex items-center gap-2">
          <BookOpen className="w-8 h-8 text-indigo-600" />
          <h1 className="text-2xl font-bold text-gray-900">English Pro</h1>
        </div>
        <div className="flex items-center gap-4">
          <select
            value={pronunciationLang}
            onChange={(e) => setPronunciationLang(e.target.value)}
            className="px-3 py-2 border border-gray-300 rounded-lg text-sm font-medium">
            <option value="en-US"><img alt="US Flag" data-bbox="468 854 486 866"/> American</option>
            <option value="en-GB"><img alt="UK Flag" data-bbox="468 873 486 885"/> British</option>
            <option value="en-AU"><img alt="AU Flag" data-bbox="468 892 486 904"/> Australian</option>
          </select>
          <Settings className="w-6 h-6 text-gray-600 cursor-pointer hover:text-gray-700"/>
        </div>
      </div>
    </div>
  </div>
);

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    </div>
  </div>
</div>

{/* Mode Toggle */}
<div className="max-w-6xl mx-auto px-4 py-6 flex gap-4">
  <button
    onClick={() => {
      setConversationMode('structure');
      setMessages([]);
    }}
    className={`flex items-center gap-2 px-6 py-3 rounded-lg font-semibold
      conversationMode === 'structure'
        ? 'bg-indigo-600 text-white shadow-lg'
        : 'bg-gray-100 text-gray-700 hover:bg-gray-200'
    }`>
    >
    <BookOpen className="w-5 h-5" />
    Structures
  </button>
  <button
    onClick={() => {
      setConversationMode('chat');
      setSelectedPhrase(null);
      setCorrection(null);
    }}
    className={`flex items-center gap-2 px-6 py-3 rounded-lg font-semibold
      conversationMode === 'chat'
        ? 'bg-indigo-600 text-white shadow-lg'
        : 'bg-gray-100 text-gray-700 hover:bg-gray-200'
    }`>
    >
    <MessageCircle className="w-5 h-5" />
    Chat
  </button>
</div>

<div className="max-w-6xl mx-auto px-4 pb-10">
  {conversationMode === 'structure' ? (
    <>
      {/* Structure Header */}
      <div className="bg-white rounded-2xl shadow-lg p-8 mb-8">
        <div className="flex items-start justify-between mb-6">
          <div>
            <div className="text-sm font-semibold text-indigo-600 mb-2">
              Structure {currentStructureIdx + 1} / {STRUCTURES.length}
            </div>
          </div>
        </div>
      </div>
    </>
  ) : (
    <div className="flex items-center justify-between mb-6">
      <div>
        <div className="text-sm font-semibold text-indigo-600 mb-2">
          Chat {currentChatIdx + 1} / {CHATS.length}
        </div>
      </div>
    </div>
  )}
</div>

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<h2 className="text-4xl font-bold text-gray-900 mb-4">
  {currentStructure.structure}
</h2>
<div className="flex items-center gap-2 mb-4">
  <button
    onClick={() => speak(currentStructure.structure)}
    className="flex items-center gap-2 px-4 py-2 bg-indigo-100"
  >
    <Volume2 className="w-5 h-5" />
    Prononcé
  </button>
</div>
</div>
<button
  onClick={() => setShowExplanation(!showExplanation)}
  className="px-4 py-2 bg-amber-100 text-amber-700 rounded-lg fon"
>
  {showExplanation ? 'Masquer expl.' : 'Voir expl.'}
</button>
</div>

{showExplanation && (
  <div className="bg-amber-50 border-l-4 border-amber-400 p-4 round"
    <p className="text-amber-900 font-semibold mb-2">🗣️ Quand l'uti
    <p className="text-amber-800 text-lg">{currentStructure.explana
  </div>
)}
</div>

{/* Examples Grid */}
<div className="grid grid-cols-1 md:grid-cols-2 gap-4 mb-8">
  {currentStructure.examples.map((example, idx) => (
    <div
      key={idx}
      onClick={() => handlePhraseSelection(idx)}
      className={`p-6 rounded-xl cursor-pointer transition-all border
        selectedPhrase === idx
          ? 'border-indigo-600 bg-indigo-50 shadow-lg'
          : 'border-gray-200 bg-white hover:border-indigo-300'
      }`}
    >
    <div className="flex justify-between items-start mb-3">
      <p className="text-lg font-semibold text-gray-900">{example}<
      <button
        onClick={(e) => {
          e.stopPropagation();
          speak(example);

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        }}
        className="p-2 bg-gray-100 hover:bg-gray-200 rounded-lg"
    >
        <Volume2 className="w-5 h-5 text-gray-600" />
    </button>
</div>
</div>
)))
</div>

{/* Practice Section */}
{selectedPhrase !== null && (
    <div className="bg-white rounded-2xl shadow-lg p-8 mb-8">
        <h3 className="text-2xl font-bold text-gray-900 mb-6">
            🎤 Répétez cette phrase :
        </h3>
        <div className="bg-indigo-50 p-6 rounded-xl mb-6 border-2 border-
            <p className="text-xl text-gray-900">{currentStructure.examples
        </div>

        <button
            onClick={startListening}
            disabled={isListening}
            className={`w-full py-4 rounded-xl text-white font-bold text-lg
                isListening
                    ? 'bg-red-500 animate-pulse'
                    : 'bg-indigo-600 hover:bg-indigo-700'
                }`>
        >
            <Mic className="w-6 h-6" />
            {isListening ? 'Écoute...' : 'Appuyez pour parler'}
        </button>

        {correction && (
            <div className={`mt-6 p-6 rounded-xl flex items-start gap-4 ${
                correction.type === 'success'
                    ? 'bg-green-50 border-2 border-green-200'
                    : 'bg-orange-50 border-2 border-orange-200'
            }`}>
                {correction.type === 'success' ? (
                    <CheckCircle2 className="w-6 h-6 text-green-600 flex-shrink
                ) : (
                    <AlertCircle className="w-6 h-6 text-orange-600 flex-shrink
                )}
            <div className="flex-1">
                <p className={`whitespace-pre-line ${
                    correction.type === 'success'

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        ? 'text-green-900 font-semibold'
        : 'text-orange-900'
    }`}>
    {correction.message}
</p>
<div className="mt-3 bg-gray-200 rounded-full h-2 overflow-
  <div
    className={`h-full transition-all ${
      correction.type === 'success'
        ? 'bg-green-500'
        : 'bg-orange-500'
    }`}
    style={{ width: `${correction.percentage}%` }}
  />
</div>
<p className="text-sm mt-2 font-semibold">
  Précision: {correction.percentage}%
</p>
</div>
</div>
  )}
</div>
)}

{/* Navigation */}
<div className="flex gap-4">
  <button
    onClick={() => {
      setCurrentStructureIdx(Math.max(0, currentStructureIdx - 1));
      setSelectedPhrase(null);
      setCorrection(null);
    }}
    disabled={currentStructureIdx === 0}
    className="flex-1 py-3 bg-gray-200 text-gray-700 rounded-lg font-
  >
    ← Précédent
  </button>
  <button
    onClick={() => {
      setCurrentStructureIdx(Math.min(STRUCTURES.length - 1, currentS
      setSelectedPhrase(null);
      setCorrection(null);
    }}
    disabled={currentStructureIdx === STRUCTURES.length - 1}
    className="flex-1 py-3 bg-indigo-600 text-white rounded-lg font-b
  >
    Suivant →

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        </button>
      </div>
    </>
  ) : (
    <>
      { /* Chat Mode */ }
      <div className="bg-white rounded-2xl shadow-lg p-8 flex flex-col h-[600px]">
        <div className="flex-1 overflow-y-auto mb-6 space-y-4">
          {messages.length === 0 ? (
            <div className="text-center py-12">
              <MessageCircle className="w-16 h-16 text-gray-300 mx-auto mb-4">
                <p className="text-gray-500 text-lg">Start a conversation in
              </div>
            ) : (
              messages.map((msg, idx) => (
                <div
                  key={idx}
                  className={`flex ${msg.role === 'user' ? 'justify-end' : 'justify-start'} space-x-4`}
                >
                  <div
                    className={`max-w-xs px-4 py-3 rounded-lg ${
                      msg.role === 'user'
                        ? 'bg-indigo-600 text-white'
                        : 'bg-gray-100 text-gray-900'
                    }`}
                  >
                    <p>{msg.content}</p>
                  </div>
                </div>
              ))
            )}
          </div>
          <div className="flex gap-3">
            <input
              type="text"
              value={userInput}
              onChange={(e) => setUserInput(e.target.value)}
              onKeyDown={(e) => e.key === 'Enter' && sendMessage()}
              placeholder="Type or speak in English..."
              className="flex-1 px-4 py-3 border border-gray-300 rounded-lg focus:outline-none"
            />
            <button
              onClick={startListening}
              className={`p-3 rounded-lg transition-all ${
                isListening
                  ? 'bg-red-500 text-white animate-pulse'

```

```

        : 'bg-gray-100 hover:bg-gray-200'
      }`}
    >
    <Mic className="w-6 h-6" />
  </button>
  <button
    onClick={sendChatMessage}
    className="px-6 py-3 bg-indigo-600 text-white rounded-lg hover:
  >
    <Send className="w-5 h-5" />
  </button>
</div>
</div>
</>
  )}
</div>
</div>
  );
};

export default EnglishLearningSaaS;

```